

Thesis

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1 Introduction

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3 Methodology

3.1 SAM-3

3.2 Prompts

For each error type, I tried to think of some fitting prompts that could catch the given errors. An important aspect when writing prompts was to include some similar ones, changing only one word, or the form of the word such as using singular or plural. One prompt was the respective name of the error type in the MVTech dataset, as that should be a fitting description. Additionally I wanted to see how well more general prompts can perform, so I also ran "damage", "damaged", "broken" and "error" for every type at first. Later on I dropped "error" as a prompt as it barely got any results at all, even compared to the other three.

3.3 Running the Model

As the first step I wrote all the prompts in a .json file containing an entry for each category and error type of the MVTech dataset. The model was then run with each prompt individually returning predicted masks, corresponding certainties and anomaly maps. Since SAM-3 only returns anomaly maps and masks if there was an entity found with a certainty as high as the given threshold, the confidence threshold was set to 0.25, as that seemed like a good tradeoff of filtering out noise and still including most relevant findings. In order to get a single mask and anomaly map per image, the found masks were joined together and anomaly maps were also joined, with the highest value per pixel being chosen. Since the evaluation also needs good pictures to check for false positives, for each run, the "good" folder of the corresponding category was run through the model. In the case of the good pictures, blank ground truth masks were created, to make the evaluation in the next step possible.

3.4 Evaluation

To compare the performance of each prompt, different metrics were calculated. While pixel-level AUROC and AUPRO scores were also noted, primarily the image-level AUROC scores were used to create graphics. For the calculation, the integrated evaluation functions in Anomalib were used.

4 Results

As expected, the results varied a lot between prompts and categories, with some performing well and showing good results, and some not showing any at all. In a few cases they even performed either worse than a random classifier would or showed unexpected behavior.

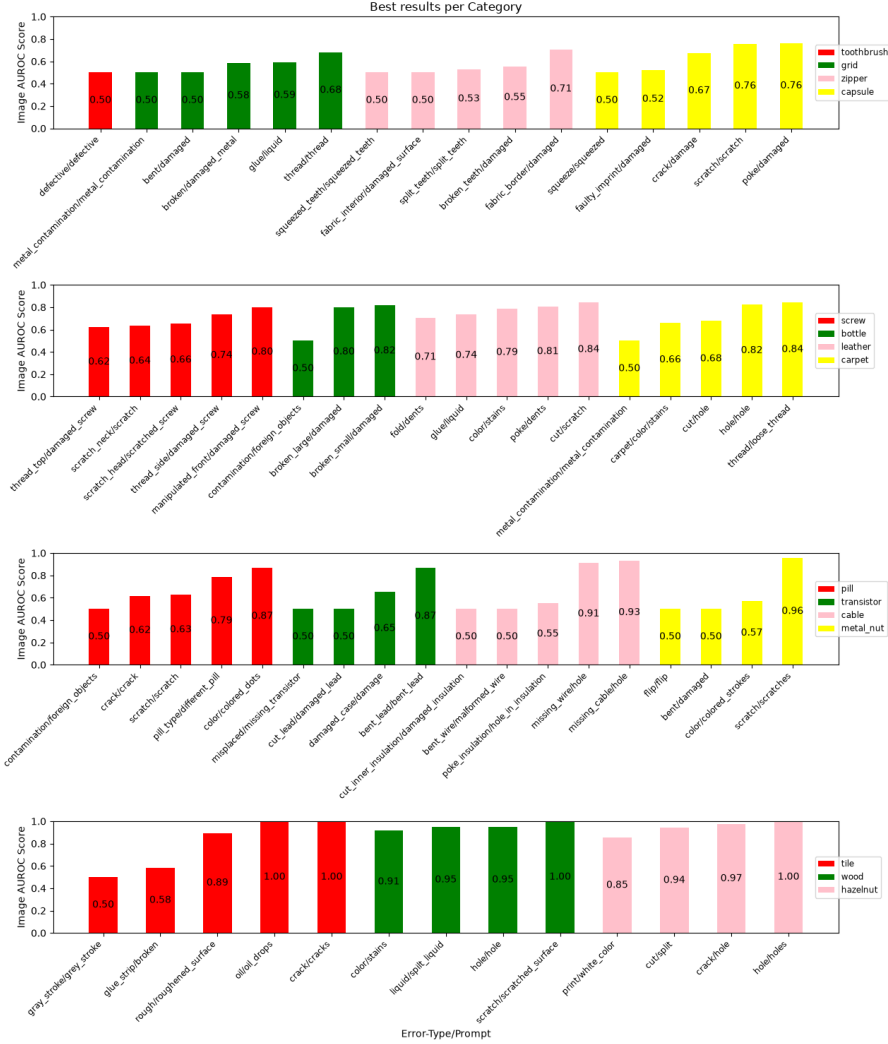


Figure 1: The 5 best image AUROC results of each category in the dataset, sorted by highest single result.

Looking at figure 1, there are four prompts that scored the best possible value of 1.0. The categories where SAM-3 performed the best overall were "hazelnut" and "wood" in which for every error type at least one prompt could score a value of 0.8 and 0.9, respectively. In the "toothbrush" category not a single prompt performed better than 0.5, meaning a random classifier would be as effective. Also "grid" and "zipper" do not have any particularly good results to show for, with most being close to 0.5. While "leather" had consistently average results, other categories such as "metal nut" or "cable" got mostly low

values, with one or two relatively high ones.

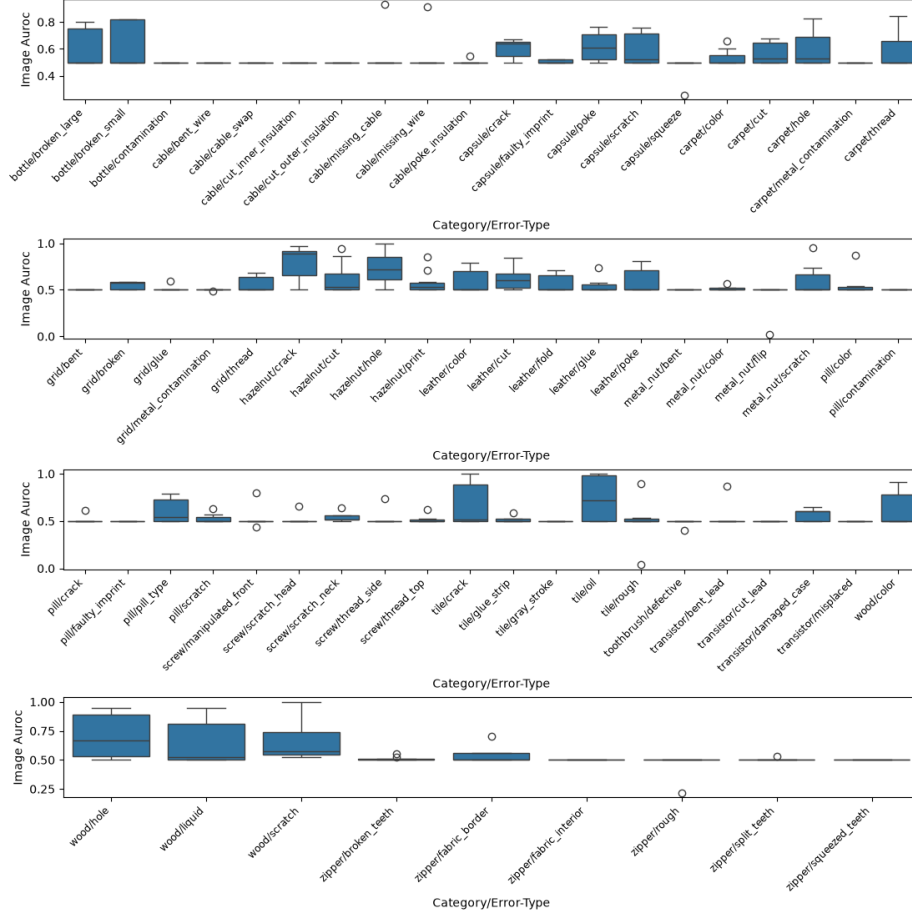


Figure 2: Boxplots of the image AUROC scores, grouped by error types.

category	error-type	number of prompts	average image auroc	best image auroc	best prompt
0	bottle	15	0.58	0.82	damaged
1	broken_large	5	0.61	0.80	damaged
2	broken_small	5	0.63	0.82	damaged
3	contamination	5	0.50	0.50	foreign_objects
4	cable	39	0.52	0.93	hole
5	bent_wire	5	0.50	0.50	malformed_wire
6	cable_swap	5	0.50	0.50	swapped_cables
7	cut_inner_insulation	5	0.50	0.50	damaged_insulation
8	cut_outer_insulation	5	0.50	0.50	damaged_insulation
9	missing_cable	6	0.57	0.93	hole
10	missing_wire	7	0.56	0.91	hole
11	poke_insulation	6	0.51	0.55	hole_in_insulation
12	capsule	27	0.56	0.76	damaged
13	crack	6	0.61	0.67	damage
14	faulty_imprint	5	0.51	0.52	damaged
15	poke	6	0.62	0.76	damaged
16	scratch	5	0.60	0.76	scratch
17	squeeze	5	0.45	0.50	squeezed
18	carpet	31	0.56	0.84	loose_thread
19	color	7	0.54	0.66	stains
20	cut	7	0.57	0.68	hole
21	hole	6	0.60	0.82	hole
22	metal_contamination	6	0.50	0.50	metal_contamination
23	thread	5	0.60	0.84	loose_thread
24	grid	28	0.52	0.68	thread
25	bent	6	0.50	0.50	damaged
26	broken	5	0.55	0.58	damaged_metal
27	glue	6	0.52	0.59	liquid
28	metal_contamination	6	0.50	0.50	metal_contamination
29	thread	5	0.56	0.68	thread
30	hazelnut	35	0.66	1.00	holes
31	crack	7	0.79	0.97	hole
32	cut	9	0.62	0.94	split
33	hole	8	0.74	1.00	holes
34	print	11	0.57	0.85	white_color
35	leather	30	0.59	0.84	scratch
36	color	6	0.59	0.79	stains
37	cut	6	0.62	0.84	scratch
38	fold	6	0.57	0.71	dents
39	glue	6	0.55	0.74	liquid
40	poke	6	0.60	0.81	dents
41	metal_nut	22	0.52	0.96	scratches
42	bent	5	0.50	0.50	damaged
43	color	5	0.52	0.57	colored_strokes
44	flip	5	0.40	0.50	flip
45	scratch	7	0.61	0.96	scratches
46	pill	35	0.54	0.87	colored_dots
47	color	6	0.57	0.87	colored_dots
48	contamination	6	0.50	0.50	foreign_objects
49	crack	5	0.52	0.62	crack
50	faulty_imprint	5	0.50	0.50	faulty_text
51	pill_type	6	0.61	0.79	different_pill
52	scratch	7	0.53	0.63	scratch
53	screw	27	0.54	0.80	damaged_screw
54	manipulated_front	5	0.55	0.80	damaged_screw
55	scratch_head	5	0.53	0.66	scratched_screw
56	scratch_neck	5	0.55	0.64	scratch
57	thread_side	6	0.54	0.74	damaged_screw
58	thread_top	6	0.52	0.62	damaged_screw
59	tile	31	0.60	1.00	oil_drops
60	crack	6	0.67	1.00	cracks
61	glue_strip	5	0.52	0.58	broken
62	gray_stroke	6	0.50	0.50	grey_stroke
63	oil	8	0.74	1.00	oil_drops
64	rough	6	0.49	0.89	roughened_surface
65	toothbrush	5	0.48	0.50	defective
66	defective	5	0.48	0.50	defective
67	transistor	20	0.54	0.87	bent_lead
68	bent_lead	5	0.57	0.87	bent_lead
69	cut_lead	5	0.50	0.50	damaged_lead
70	damaged_case	5	0.57	0.65	damage
71	misplaced	5	0.50	0.50	missing_transistor
72	wood	23	0.67	1.00	scratched_surface
73	color	6	0.63	0.91	stains
74	hole	6	0.71	0.95	hole
75	liquid	6	0.65	0.95	spill_liquid
76	scratch	5	0.68	1.00	scratched_surface
77	zipper	38	0.50	0.71	damaged
78	broken_teeth	8	0.51	0.55	damaged
79	fabric_border	5	0.55	0.71	damaged
80	fabric_interior	5	0.50	0.50	damaged_surface
81	rough	7	0.46	0.50	rough_tooth
82	split_teeth	7	0.50	0.53	split_teeth
83	squeezed_teeth	6	0.50	0.50	squeezed_teeth

Figure 3: Number of different prompts used per category and error type together with the best prompt for each.

4.1 Pitfalls

A high image AUROC score does not always mean that the model is actually doing what we want it to. One example for this in my experiment is the error type "scratch" in the category "wood" with the prompt "scratched surface". While the calculated image AUROC score is 1.0, the highest possible value, after looking at the predicted masks, it becomes clear that the prompt does not actually find the defects I was looking for.

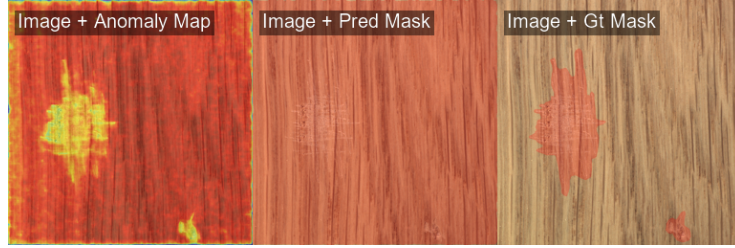


Figure 4: Example image from the dataset together with the anomaly map, predicted masks and ground truth masks, respectively.

Here it can be clearly seen that this is not the wanted result where the model only marks the parts with scratches, but rather the whole image. In the anomaly map, the scratched parts are even given a lower confidence. This gives the impression that the prompt was not interpreted as intended and focused more on the "surface" part, ignoring the "scratched" before that.

5 Discussion

Overall it can be said, that using SAM-3 with text prompts for anomaly detection can work, however it still has some clear weaknesses and shortcomings that would need to be further worked on. Judging by the results of my experiments, this can be a helpful tool in addition to other methods or an alternative if a zero shot model is needed.